

41. What is a Python function?

Answer:

A function is a block of reusable code that performs a specific task. Functions help to avoid writing the same code again and again.

Example:

```
def greet():  
    print("Hello Python")
```

```
greet()
```

Output:

Hello Python

42. What are the advantages of using functions in Python?

Answer:

Advantages of functions:

1. **Code reusability** – Write code once and use it many times.
2. **Easy debugging** – Errors can be found easily.
3. **Better organization** – Makes programs clean and readable.
4. **Reduces code duplication.**

43. What is the difference between a parameter and an argument?

Answer:

Parameter:

A variable used when creating a function.

Argument:

The actual value passed to the function when calling it.

Example:

```
def add(a, b): # a and b are parameters  
    print(a + b)
```

```
add(10, 20)  # 10 and 20 are arguments
```

44. What is a return statement in Python?

Answer:

The return statement sends a value back from a function.

Example:

```
def add(a, b):  
    return a + b
```

```
result = add(5, 3)  
print(result)
```

Output:

8

45. What are default arguments in Python?

Answer:

Default arguments are values given to function parameters during function definition. If no value is passed, the default value is used.

Example:

```
def greet(name="User"):  
    print("Hello", name)
```

```
greet()  
greet("Bhagya")
```

Output:

Hello User

Hello Bhagya